

UNIT : 2

* Simple Transformation

* Taylor Series

* Laurent's Series

* Bilinear

* Conformal Mapping

* Singularities

Transformation

1. Discuss a transformation $w = C + z$

Solu

$w = C + z$ (Translation) where C is

complex constant

w.k.T

$$w = u + iv$$

$$z = x + iy$$

assume that $c = a + ib$

$$w = c + z \rightarrow \textcircled{1} \text{ (Translation)}$$

$$\textcircled{1} \Rightarrow u + iv = (a + ib) + (x + iy)$$

$$u + iv = (a + x) + i(b + y)$$

Separate real & Imag.

$$u = x + a \quad \& \quad v = y + b$$

The pt $P(x, y)$ in z plane is
mapped on to the pt $P'(x+a, y+b)$
Hence the two region have the same
shapes

2. $w = cz$ (Magnification or Rotation)

A.T

where w, z, c are complex number

number

Let $z = re^{i\alpha}$

$w = Re^{i\phi}$

$c = ae^{i\alpha}$

$w = cz$

$Re^{i\phi} = ae^{i\alpha} re^{i\alpha}$

$\Rightarrow Re^{i\phi} = ar e^{i(\alpha+\theta)}$

$Re^{i\phi} = ar e^{i(\alpha+\theta)}$

Separate real & Imag

$R = ar$

$e^{i\phi} = e^{i(\alpha+\theta)}$

$= \left(\frac{r}{x_0} \right) + i \left(\frac{y}{x_0} \right)$

Then a point (x, y) in z plane
is mapped on to the pt (u, v)

(3) $w = e^z$

G.T

$$w = e^z$$

$$u + iv = e^{x+iy}$$

$$= e^x e^{iy}$$

$$= e^x (\cos y + i \sin y)$$

$$u + iv = e^x \cos y + i e^x \sin y$$

Separate real & Imag. part

$$u = e^x \cos y \rightarrow \textcircled{1}$$

$$u = e^x \cos y \rightarrow \textcircled{1}$$

$$v = e^x \sin y \rightarrow \textcircled{2}$$

Case (i): Eliminate y in equ $\textcircled{1}$ & $\textcircled{2}$

$$u = e^x \cos y \quad v = e^x \sin y$$

$$\frac{u}{e^x} = \cos y \quad \& \quad \frac{v}{e^x} = \sin y$$

Squaring & adding

$$\left(\frac{u}{e^x}\right)^2 + \left(\frac{v}{e^x}\right)^2 = \cos^2 y + \sin^2 y$$
$$= 1$$

$$\Rightarrow \frac{u^2}{e^{2x}} + \frac{v^2}{e^{2x}} = 1$$

$$\Rightarrow \frac{u^2 + v^2}{e^{2x}} = 1$$

$$u^2 + v^2 = e^{2x}$$

Equation of line parallel to y axis

(i.e) $x = 0$

$$u^2 + v^2 = e^0 = 1$$

$$u^2 + v^2 = 1$$

$$(u-0)^2 + (v-0)^2 = 1$$

which gives the unit circle with center at origin in w plane.

Case (ii) : Eliminate 'x' in equ (1) & (2)

$$\frac{u}{\cos y} = e^x \quad \& \quad \frac{v}{\sin y} = e^x$$

$$\frac{v}{\sin y} = \frac{u}{\cos y}$$

$$\frac{v}{u} = \frac{\sin y}{\cos y}$$

$$\frac{v}{u} = \tan y$$

$$v = u \tan y$$

Line parallel to x -axis (i.e) $y=0$

$$v = u \tan(\theta)$$

$$\boxed{v=0}$$

Hence real axis in z plane is transformed into real axis in w plane

④ ✓ $w = z + \frac{1}{z}$

$$u+iv = re^{i\theta} + \frac{1}{re^{i\theta}}$$

$$= re^{i\theta} + \frac{1}{r} e^{-i\theta}$$

$$= r(\cos\theta + i\sin\theta) + \frac{1}{r}(\cos\theta - i\sin\theta)$$

$$\textcircled{9} \textcircled{1} = r\cos\theta + i r\sin\theta + \frac{1}{r}\cos\theta - \frac{i}{r}\sin\theta$$

$$u+iv = \left(r + \frac{1}{r}\right)\cos\theta + i\left(r - \frac{1}{r}\right)\sin\theta$$

$$= r (\cos \theta + i \sin \theta) + \frac{1}{r} (\cos \theta - i \sin \theta)$$

$$\textcircled{2} \times \textcircled{1} = r \cos \theta + i r \sin \theta + \frac{1}{r} \cos \theta - \frac{i}{r} \sin \theta$$

$$u + iv = \left(r + \frac{1}{r}\right) \cos \theta + i \left(r - \frac{1}{r}\right) \sin \theta$$

$$u = \left(r + \frac{1}{r}\right) \cos \theta \quad \rightarrow \textcircled{1}$$

$$v = \left(r - \frac{1}{r}\right) \sin \theta \quad \rightarrow \textcircled{2}$$

Case (i): Eliminate 'θ' equ $\textcircled{1}$ & $\textcircled{2}$

$$\frac{u}{r + \frac{1}{r}} = \cos \theta$$

$$\frac{v}{r - \frac{1}{r}} = \sin \theta$$

Squaring and adding

$$\left(\frac{u}{r + \frac{1}{r}}\right)^2 + \left(\frac{v}{r - \frac{1}{r}}\right)^2 = 1 \cos^2 \theta + \sin^2 \theta$$

$$= 1 \frac{u^2}{(r + \frac{1}{r})^2} + \frac{v^2}{(r - \frac{1}{r})^2}$$

$$\frac{u^2}{\left(r + \frac{1}{r}\right)^2} + \frac{v^2}{\left(r - \frac{1}{r}\right)^2} = 1$$

$$\frac{u^2}{a^2} + \frac{v^2}{b^2} = 1$$

where

$$a = r + \frac{1}{r}$$

$$b = r - \frac{1}{r}$$

which is the required equ of the ellipse in w plane

Case (ii) Eliminate 'x' in equ ① ② ③

$$\frac{u}{\cos \theta} = \left(x + \frac{1}{x} \right) + \left(\frac{v}{\sin \theta} \right) = x - \frac{1}{x}$$

Squaring and subtracting

$$\left(\frac{u}{\cos \theta} \right)^2 - \left(\frac{v}{\sin \theta} \right)^2 = \left(x + \frac{1}{x} \right)^2 - \left(x - \frac{1}{x} \right)^2$$

$$= \cancel{x^2} + \cancel{\frac{1}{x^2}} + 2x \cdot \frac{1}{x} - \left(\cancel{x^2} - \cancel{\frac{1}{x^2}} + 2x \cdot \frac{1}{x} \right)$$

$$= 4$$

7.4 4

$$\frac{u^2}{4 \cos^2 \theta} - \frac{v^2}{4 \sin^2 \theta} = 1 \quad = \left(\frac{u}{\frac{1}{\lambda} - \lambda} \right)^2 - \left(\frac{v}{\frac{1}{\lambda} + \lambda} \right)^2$$

$$\frac{u^2}{(2 \cos \theta)^2} - \frac{v^2}{(2 \sin \theta)^2} = 1$$

$$\frac{u^2}{a^2} - \frac{v^2}{b^2} = 1 \quad = \left(\frac{u}{\frac{1}{\lambda} - \lambda} \right)^2 - \left(\frac{v}{\frac{1}{\lambda} + \lambda} \right)^2$$

where $a = 2 \cos \theta$
 $b = 2 \sin \theta$

which is the required equation of hyperbola in w -plane

⑤

$$w = z^2$$

$$u + i\bar{v} = (x + iy)^2$$

$$= x^2 + i^2 y^2 + 2ixy$$

$$u + i\bar{v} = (x^2 - y^2) + i(2xy)$$

Separate real & Imag. parts

$$\left(\frac{1}{x} - R\right) u = x^2 - y^2 \rightarrow \textcircled{1}$$

$$v = 2xy \rightarrow \textcircled{2}$$

case (i) : $x = a$ (parallel to y -axis) in eqn (2) & (3)

$$u = a^2 - y^2$$

$$v = 2ay$$

$$u = \left(\frac{v}{2a} \right)^2 - y^2$$

$$y = \frac{v}{2a}$$

$$u = \frac{v^2}{4a^2} - y^2$$

$$\frac{v^2}{4a^2} = u + y^2$$

$$u = a^2 - \left(\frac{v}{2a} \right)^2$$

$$u = a^2 - \frac{v^2}{4a^2}$$

$$\frac{v^2}{4a^2} = a^2 - u$$

$$v^2 = 4a^2(a^2 - u)$$

$$y^2 = 4ax$$

$$v^2 = 4a^2(a^2 - u)$$

The eqn represent the family of parabola

case (ii) $y = b$ (parallel to x -axis) in eqn ① & ②

Substitute $u = x^2 - b^2$ or $v = 2xb$

$$u = x^2 - \left(\frac{v}{2x}\right)^2$$

$$u = x^2 - \frac{v^2}{4x^2}$$

$$b = \frac{v}{2x}$$

$$u = x^2 - b^2 \quad x = \frac{v}{2b}$$

$$u = \left(\frac{v}{2b} \right)^2 - b^2$$

$$u = \frac{v^2}{4b^2} - b^2$$

$$\frac{v^2}{4b^2} = u + b^2$$

$$v^2 = 4b^2(u + b^2)$$

The above equ represent family of Parabola

⑥

$$w = \sin z$$

$$u + iv = \sin(x + iy)$$

$$\sin(A+B) = \sin A \cos B + \cos A \sin B$$

$$= \sin x \cos(iy) + \cos x \sin(iy)$$
$$\frac{\sin(i\theta)}{\cos(i\theta)} = \frac{i \sinh \theta}{\cosh \theta}$$

$$u + iv = \sin x \cosh y + i \cos x \sinh y$$

Separate real & Imag.

$$u = \sin x \cosh y \rightarrow (1)$$

$$v = \cos x \sinh y \rightarrow (2)$$

Case (i) : $x = a$ (line to y-axis) in equation

$$u = \sin a \cosh y$$

$$v = \cos a \sinh y$$